

Hospital Management System

Abstract

The Hospital Management System is a web-based application designed to automate and streamline hospital operations related to user registration, authentication, and role-based data management. This system enables hospitals to efficiently manage users, doctors, and patients using a digital platform, reducing manual work and improving data accuracy. The application is developed using Java Full Stack technologies such as Spring Boot for the backend, React (Vite) for the frontend, and MySQL for database storage, demonstrating real-world full-stack application development in the healthcare domain.

Existing System

In the existing system, hospital operations are largely managed through manual and paper-based processes. User registration, authentication, and patient-doctor data are recorded manually, leading to inefficiency, data inconsistency, and lack of centralized access. There is no structured system to manage roles and permissions, resulting in poor security and maintainability.

Proposed System

The proposed Hospital Management System automates hospital operations through a centralized web application. The system provides secure user registration and role-based authentication, with separate modules for users, doctors, and patients. Authentication details are stored in a centralized User module, while role-specific information is managed in Doctor and Patient modules, minimizing data redundancy and enhancing security. The system uses RESTful APIs for frontend-backend communication and CORS configuration for seamless integration. It is scalable and can be extended with features like appointment scheduling, billing, prescriptions, dashboards, and JWT-based authentication in the future.

Modules

- User Registration & Login Module
- Doctor Module
- Patient Module
- Role-based Data Management Module
- Authentication & Authorization Module
- RESTful API Integration Module
- CORS Configuration Module

Software Requirements

- Operating System: Windows 10 / Linux
- Frontend Technologies: React (Vite), HTML, CSS, JavaScript
- Backend Technology: Java, Spring Boot, REST APIs

- Database: MySQL
- Development Tools: Eclipse, Visual Studio Code, Maven, npm
- Web Server: Apache Tomcat
- Browser: Google Chrome / Mozilla Firefox

Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Hard Disk: Minimum 250 GB
- Internet Connection: Required
- Input Devices: Keyboard, Mouse
- Output Device: Monitor